



SRS

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Oral Cancer Screening & Lesion Comparison System

Platform: Web Application + Mobile Application

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the **Oral Cancer Screening & Lesion Comparison Tool**, a digital health support system designed to:

1. Assess oral cancer risk using a structured questionnaire.
2. Allow users to upload oral lesion images and retrieve visually similar benign and malignant examples for comparison (CBIR-based retrieval).

The system is **not a diagnostic tool**. It is intended for early awareness, triage guidance, and prompting timely clinical consultation.

1.2 Scope

The system will:

- Be available as a responsive **web application** and **mobile application** (Android/iOS).
- Provide a structured risk questionnaire.
- Dynamically evaluate risk using rule-based logic.
- Accept an image upload from the user.
- Retrieve the closest visually similar benign and malignant lesion images from a pre-indexed dataset using deep feature embeddings.
- Provide appropriate guidance messaging based on risk category.

The system does **not**:

- Provide a medical diagnosis.
 - Replace clinical evaluation.
 - Perform automated cancer detection classification (only similarity comparison).
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1.3 Definitions

- **Cause:** A reported risk factor (Section A).
 - **Symptom:** A reported clinical sign (Section B).
 - **Differential:** Follow-up responses suggesting alternative benign explanations.
 - **CBIR:** Content-Based Image Retrieval.
 - **Embedding:** Feature vector extracted from an image using a CNN model.
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2. Overall System Description

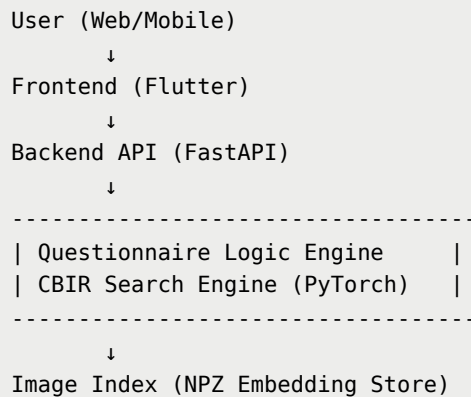
2.1 System Architecture

The system consists of two major modules:

1. Questionnaire Engine (Rule-based logic)

2. Lesion Image Comparison Engine (CBIR-based retrieval)

High-Level Architecture



3. Functional Requirements

MODULE 1: ORAL CANCER RISK QUESTIONNAIRE

3.1 Questionnaire Structure

The system shall include:

- Section A: 9 Risk Factor questions (Yes/No)
- Section B: 9 Symptom questions (Yes/No)
- 8 Conditional Differential questions (D1–D8)

3.2 Conditional Logic

The system shall:

1. Display D1–D3 only if B1 = Yes
2. Display D4–D6 only if B2 = Yes
3. Display D7–D8 only if B3 = Yes
4. Clear differential responses if parent question is changed to No

5. Dynamically update progress bar based on visible questions

3.3 Risk Evaluation Logic

The system shall evaluate three binary flags:

- CAUSE_PRESENT
- SYMPTOM_PRESENT
- DIFFERENTIAL_PRESENT

Decision Matrix

Cause	Symptom	Differential	Result
Yes	Yes	Any	Result 1 – High Risk
Yes	No	N/A	Result 2 – Elevated Risk
No	Yes	No	Result 3 – Consultation
No	Yes	Yes	Result 4 – Alternate Dx
No	No	N/A	Result 5 – No Immediate Concern

3.4 Result Output

The system shall display:

- Risk category title
 - Explanation message
 - Recommended actions list
 - Optional contextual note (if differential present in Result 1)
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3.5 Reset Behavior

Upon clicking "Start Over":

- All answers are cleared
 - Differential blocks hidden
 - Result card removed
 - Progress bar reset
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MODULE 2: LESION IMAGE COMPARISON (CBIR SYSTEM)

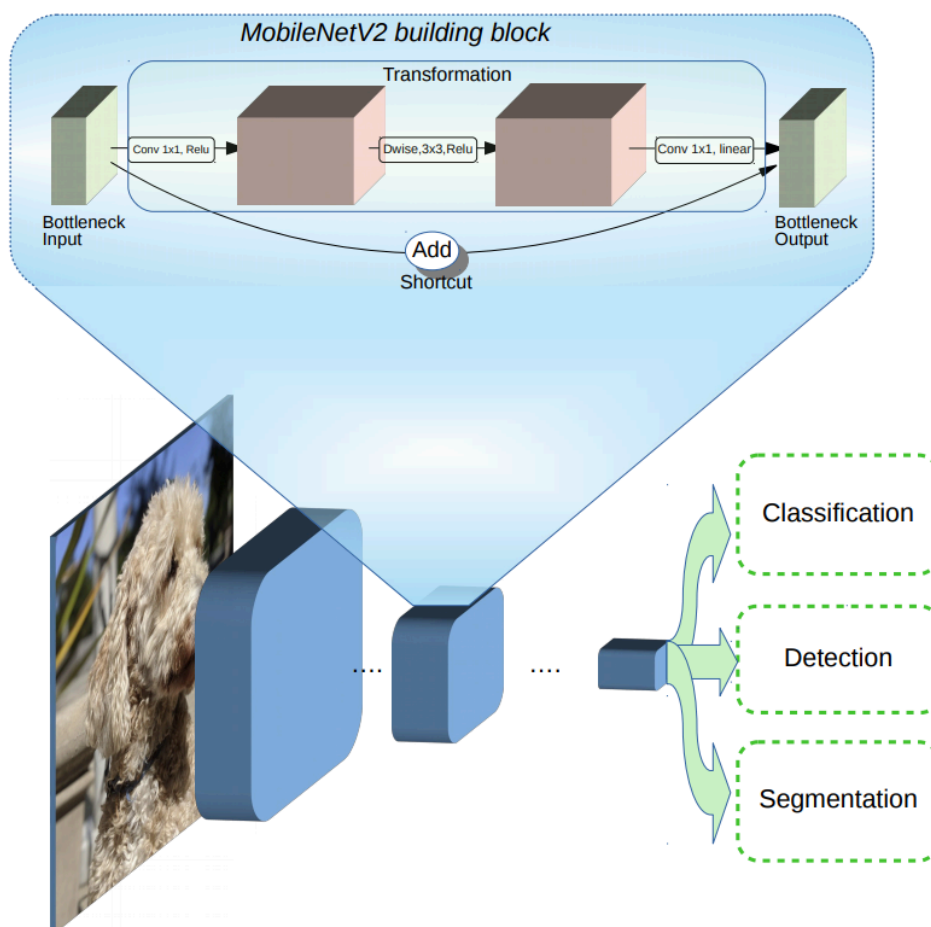
4. Overview

The second module allows users to:

1. Upload an oral cavity image.
2. Extract deep CNN embeddings.
3. Retrieve top-K most similar benign and malignant lesions.
4. Display them side-by-side for comparison.

This is a similarity retrieval system — not a classifier.

4.1 Feature Extraction



The system shall:

- Use MobileNetV2 pretrained on ImageNet.
 - Remove final classification layer.
 - Apply adaptive average pooling.
 - Extract 1280-dimensional embeddings.
 - L2-normalize embeddings.
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4.2 Index Builder

The system shall:

- Recursively scan dataset directories.
- Detect benign and malignant folder names.
- Extract embeddings for all supported image formats.
- Store embeddings in a compressed index file (cbir_index.npz).

The index shall store:

- embeddings
 - file paths
 - labels (benign/malignant)
 - source folder names
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4.3 Image Upload Requirements

The system shall:

- Accept JPG, PNG, JPEG, BMP, WEBP, TIFF formats.
 - Resize images to 224×224.
 - Convert to RGB.
 - Normalize using ImageNet mean/std.
 - Reject images > predefined size limit (e.g., 10MB).
 - Reject unsupported file formats.
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4.4 Similarity Search

The system shall:

- Compute cosine similarity between query embedding and dataset embeddings.
 - Retrieve:
 - Top K benign matches
 - Top K malignant matches
 - Display them in separate labeled sections.
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4.5 Display Requirements

The UI shall:

- Display user-uploaded image.
 - Display top benign matches.
 - Display top malignant matches.
 - Show similarity score (optional percentage).
 - Include disclaimer that similarity $\neq$ diagnosis.
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5. Non-Functional Requirements

5.1 Performance

- Questionnaire evaluation: < 1 second
 - Image retrieval response: < 3 seconds (CPU), < 1.5 seconds (GPU)
 - Index loading time: < 5 seconds
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5.2 Privacy & Compliance

- GDPR-compliant data handling.
 - Clear consent statement before image upload.
 - Explicit statement that tool is not a medical device unless certified.
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5.4 Usability

- Responsive UI for desktop and mobile.
 - Accessible font size.
 - High contrast UI.
 - Clear medical disclaimers.
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6. External Interface Requirements

- Web: React-based SPA
 - Mobile: Flutter or React Native
 - Progress bar for questionnaire
 - Clean card-based results
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7. Constraints

- Model limited to MobileNetV2 unless retraining approved.
 - Dataset quality affects retrieval accuracy.
 - System does not replace biopsy or histopathology.
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8. Assumptions

- Users can take reasonably clear intraoral photographs.
 - Dataset contains labeled benign and malignant lesions.
 - Clinical guidance text reviewed by medical professionals.
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Disclaimer

This system is intended for **educational and preliminary screening purposes only**.

It is **not a diagnostic medical device**.

Definitive diagnosis of oral cancer requires clinical examination and biopsy by a qualified healthcare professional.
